EIN 4343 Inventory and Supply Chain Systems

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A blueprint going forward

- Modeling of demand
  - Deterministic vs. stochastic

- Planning horizon length
  - Infinite vs. finite

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: STOCHASTIC DEMAND FINITE HORIZON

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Newsvendor Model

- The demand is a random variable (r.v.).
  - A variable whose values represent the possible outcomes of a “yet-to-be-performed” experiment

- Upon the realization of demand, we incur
  - either excess stock cost if the demand is less than the amount in inventory,
  - or shortage cost if the demand is more than the amount in inventory.

- Therefore, we need a different model to decide how much to purchase (or produce).

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Newsvendor assumptions

1. The inventory level is reviewed on a periodic basis (at the end of the planning period).
2. There is a single planning period.
3. The demand is random, and luckily the underlying probability distribution for the demand is known.
4. The entire order quantity can be delivered at the beginning of the planning period.
5. The order quantity can be a continuous or an integer value depending on the type of r.v. used to characterize the demand.
   - A continuous r.v. takes values from an infinite set, i.e., a continuum.
   - A discrete r.v. takes values from a finite set.

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Newsvendor assumptions

6. The system under consideration is a single item system.
7. The cost and revenue parameters for the period are known.
8. There is no fixed cost associated with placing an order.
9. The order delivery lead-time is zero.
10. Shortages are allowed.

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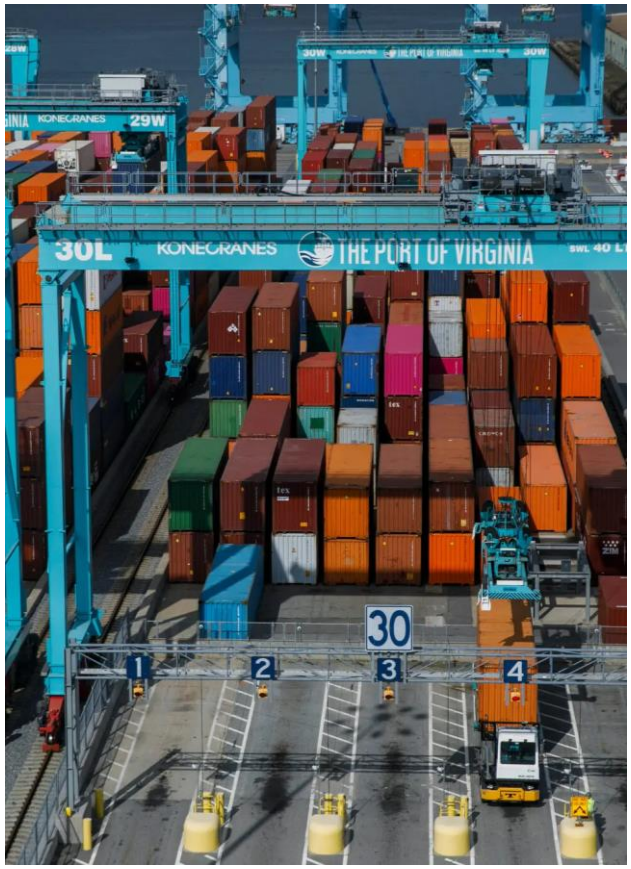
2024: Longshoreman strike at East and Gulf Coast ports…

- *…There is no practical alternative to the East and Gulf Coast ports for moving many goods in and out of the eastern half of the country. And the ports can't operate without longshoremen, who move metal boxes called containers on and off ships and handle other economically crucial cargo, like cars and heavy machinery.*
- *Fearing a strike, many businesses ordered goods early so that they cleared the East and Gulf Coast ports before Tuesday. But much cargo didn't make it through in time and could be stranded at sea. Even if a strike is short, it could take weeks to clear the backlog.*
- *John Wrenn, the chief operating officer at MHW, an alcoholic beverages distributor based in Manhasset, N.Y., is concerned that a strike might delay shipments and leave his business short in the lead up to Thanksgiving.*
- *"Those are big opportunities for sales that will just be lost because products will not be on the shelves," he said. MHW imports over 2,000 containers a year through the Port of New York and New Jersey, the third busiest port in the United States.*
- *Stefanie Katzman, executive vice president at S. Katzman Produce, a fruit and vegetable supplier in the Bronx, is particularly concerned that mangos, which are imported from Brazil, will spoil if they are stuck on a ship. She is considering flying some mangos in, a much more expensive option. "There's nowhere near as much room to fly product as there is to send it in by boat," Ms. Katzman said.*



https://www.nytimes.com/2024/09/30/business/port-strike-dockworkers.html

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Newsvendor: Notation

X	random variable that represents the demand in the period;
$f(x)$	probability density function of X over the support $[0, \infty)$;
$\mu = E[X]$	mean of X;
$\sigma^2 = \mathrm{Var}(X)$	variance of X;
$F(x)$	cumulative density function of X;
c_o	unit cost of overage (\$/unit);
c_u	unit cost of underage (\$/unit);
q	quantity ordered prior to the beginning of the period(units);
$c(q, X)$	total cost at the end of the period (\$);
$C(q) = E_X\,[c(q, X)]$	the expected cost function (\$).

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Newsvendor: Setting

- Minimize the expected total cost at the end of the planning period.
- Inventory on hand prior to the beginning of the period is q.
- Demand observed by the end of the period is X.

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Newsvendor: Intuition

- Let's play a game
- Click on this link and enter your initials and a value in the region [30,70]
- https://uflorida-my.sharepoint.com/:x:/r/personal/akcali_ufl_edu/Documents/S26%20Newsvendor%20Game.xlsx?d=w8bc4e787ae0442a79c3665996a0fc88b&csf=1&web=1&e=jmrhKY
- That is, let us generate a set of outcomes from a continuous uniform distribution between 30 and 70.

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Newsvendor: Costs

At the end of the period, we incur

- **either** a unit cost of c_o for each unit that is left over in inventory, i.e., <u>overage cost</u>,

- **or** a unit cost of c_u for each unit of unsatisfied demand, i.e., <u>underage cost</u>.

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Newsvendor: Model setup

What is the realized cost if the order quantity is q and the random demand is X?

$$c(q,X) = \begin{cases} c_o(q-X) & X \le q \quad (1) \\ c_u(X-q) & q < X \quad (2) \end{cases}$$

What is the expected total cost over all possible values of X?

$$E_X[c(q,X)] = \underbrace{\boxed{\int_0^q c_o(q-X)f(x)dx}}_{(1)} + \underbrace{\boxed{\int_q^\infty c_u(X-q)f(x)dx}}_{(2)} = C(q)$$

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Newsvendor: Model setup

$$C(q) = \int_0^q c_o(q - X)f(x)dx + \int_q^\infty c_u(X - q)f(x)dx$$

- This is a single variable unconstrained optimization problem.

- To determine the optimal order quantity, we need to examine the first and second order conditions.

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Controversy of Calculus

Who invented calculus?

Isaac Newton

Gottfried Wilhem Leibnitz

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Newsvendor: Leibniz's rule

For differentiation of integrals, we have the following:

$$\frac{d}{dy}\int_{\ell(y)}^{u(y)} h(x,y)dx = \int_{\ell(y)}^{u(y)} \frac{\partial h(x,y)}{\partial y} dx + h(u(y),y)u'(y)$$

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Probability Refresher

- We have

$$\int_0^\infty f(x)dx = 1$$

- Also, we have

$$\int_0^q f(x)dx = F(q)$$

$$\int_q^\infty f(x)dx = \int_0^\infty f(x)dx - \int_0^q f(x)dx = 1 - \int_0^q f(x)dx = 1 - F(q)$$

- Finally, we also have

$$\int_0^\infty xf(x)dx = E[X] = \mu_X$$

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Newsvendor: First order condition

$$C(q) = \int_0^q c_o(q - X)f(x)dx + \int_q^\infty c_u(X - q)f(x)dx$$

$$\frac{dC(q)}{dq} = c_o\left[\int_0^q (1)f(x)dx + (q - q)(1) - (q - 0)(0)\right]$$

$$+c_u\left[\int_q^\infty (-1)f(x)dx + (\infty - q)(0) - (q - q)(1)\right]$$

$$= c_o\int_0^q f(x)dx - c_u\int_q^\infty f(x)dx$$

$$= c_o\int_0^q f(x)dx - c_u\left[\int_0^\infty f(x)dx - \int_0^q f(x)dx\right]$$

$$= c_oF(q) - c_u[1 - F(q)]$$

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Newsvendor: Second order condition

$$C(q) = \int_0^q c_o(q - X)f(x)dx + \int_q^\infty c_u(X - q)f(x)dx$$

$$\frac{dC(q)}{dq} = c_o F(q) - c_u[1 - F(q)]$$

$$\frac{d^2C(q)}{q^2} = c_o f(x) + c_u f(x) > 0$$

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Newsvendor: Optimal order quantity

$$C(q) = \int_0^q c_o(q - X)f(x)dx + \int_q^\infty c_u(X - q)f(x)dx$$

$$\frac{dC(q)}{dq} = c_o F(q) - c_u[1 - F(q)] = 0$$

$$F(q^*) = \frac{c_u}{c_u + c_o}$$

Note that $\frac{c_u}{c_u + c_o}$ is known as the critical ratio.

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Newsvendor: Billy’s Bagels

Billy's Bagels bakes fresh bagels each morning. The daily demand for bagels can be characterized using a discrete random variable for which an empirical distribution can be obtained using historical sales data. The empirical distribution is as given in the Figure.

Dozens of Bagels sold	*Probability of Occurrence*
0	.05
5	.10
10	.10
15	.20
20	.25
25	.15
30	.10
35	.05

The bagels cost 8 cents/unit to bake and are sold for 35 cents/unit. Bagels unsold at the end of the day are purchased by a nearby charity soup kitchen for 3 cents/unit.

How many dozens of bagels should Billy’s bake each day?

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Newsvendor: Billy's Bagels

$cost = \$0.08/\text{unit}$
$selling\ price = \$0.35/\text{unit}$
$salvage\ revenue = \$0.03/\text{unit}$

- Unit underage cost

$$c_u = 0.35 - 0.08 = \$0.27/\text{unit}$$

- Unit overage cost

$$c_o = 0.08 - 0.03 = \$0.05/\text{unit}$$

- Critical ratio

$$F(q^*) = \frac{c_u}{c_u + c_0} = \frac{0.27}{0.27 + 0.05} = 0.8438$$

Note that the unit underage and overage costs are evaluated per unit as opposed to per dozen – which would have no impact on the optimal solution for this problem context.

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Newsvendor: Billy's Bagels

Dozens of Bagels Sold	Probability	Cumulative Probability
0	0.05	0.05
5	0.10	0.15
10	0.10	0.25
15	0.20	0.45
20	0.25	0.70
25	0.15	0.85
30	0.10	0.95
35	0.05	1.00

$F(q^*) = 0.8438 \quad \rightarrow q^* = 25 \text{ dozens of bagels}$

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Newsvendor: St Valentine's Cards

A drugstore in Coral Springs, Florida sells St. Valentine's Day cards.

The store manager has been keeping track of the sales volume and he has the following empirical demand distribution for the sales volume where the quantity is uniformly distributed within each range.

The cards cost 99 cents/unit to buy and are sold for $1.79 /unit. Cards unsold by St Valentine's day can neither be sold at a discounted rate nor be stored until the next year.

How many cards should this store order for next year?

Quantity Sold	*Probability of Occurrence*
100–150	*.10*
150–200	*.15*
200–250	*.25*
250–300	*.30*
300–350	*.15*
> 350	*.05*

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Newsvendor: St. Valentine's cards

$cost = \$0.99/unit$
$selling\ price = \$1.79/unit$
$salvage\ revenue = \$0.00/unit$

- Unit underage cost

$$c_u = 1.79 - 0.99 = 0.80/card$$

- Unit overage cost

$$c_o = 0.99 = 0.99/card$$

- Critical ratio

$$F(q^*) = \frac{c_u}{c_u + c_0} = \frac{0.80}{0.80 + 0.99} = 0.4469$$

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Note

- Recall linear interpolation. If the two known points are given by the coordinates (x_0, y_0) and (x_1, y_1), the linear interpolant is the straight line between these points.

- For a value x in the interval (x_0, x_1), the value y along the straight line is given from the equation

$$\frac{y - y_0}{x - x_0} = \frac{y_1 - y_0}{x_1 - x_0}$$

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Newsvendor: St. Valentine's cards

Quantity sold	Probability	Cumulative Probability
100-150	0.10	0.10
150-200	0.15	0.25
200-250	0.25	0.50
250-300	0.30	0.80
300-350	0.15	0.95
350-	0.05	1.00

$F(q^*) = 0.4469$

Recall that demand can assumed to be uniformly distributed within each range. Hence, we have

$$\frac{F(q^*) - 0.25}{0.50 - 0.25} = \frac{q^* - 200}{250 - 200} \rightarrow q^* = 239.38\ cards$$

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Newsvendor: Bread

A supermarket must decide how many loaves of a particular type of whole wheat bread to purchase every day. The daily demand for this type of bread is approximately normally distributed with a mean of 300 and a standard deviation of 50 loaves.

A loaf of bread sells for $2.19 and costs the supermarket $1.59.

A loaf of bread that is not sold by the end of the day can be sold as a *"day old"* item the next day for $0.99.

How many loaves of this type of whole wheat bread should the supermarket purchase every day?

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Newsvendor: Bread

$cost = \$1.59/\text{unit}$
$selling\ price = \$2.19/\text{unit}$
$salvage\ revenue = \$0.99/\text{unit}$

- Unit underage cost

$$c_u = 2.19 - 1.59 = 0.60/\text{unit}$$

- Unit overage cost

$$c_o = 1.59 - 0.99 = 0.60/\text{unit}$$

- Critical ratio

$$F(q^*) = \frac{c_u}{c_u + c_0} = \frac{0.6}{0.6 + 0.6} = 0.5$$

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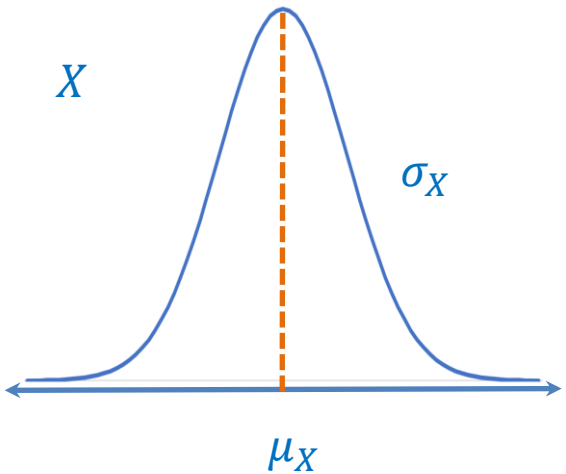
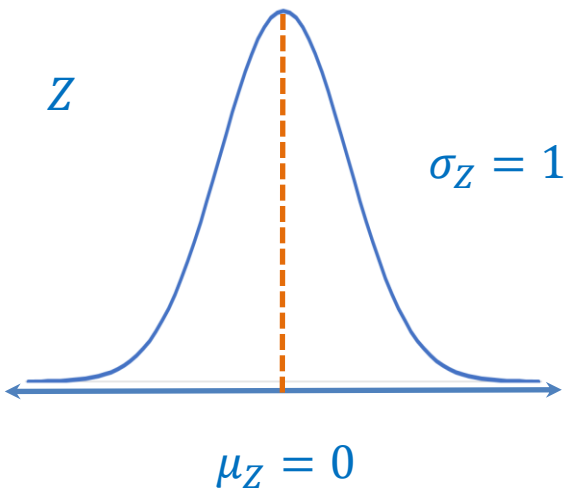
Probability refresher





$$P\{X < x\} = P\left\{\frac{X - \mu_X}{\sigma_X} < \frac{x - \mu_X}{\sigma_X}\right\} = P\{Z < z\}$$

$$F(x) = \Phi(z)$$

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Z-table

Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)
-3.00	0.0013	3.000	-1.48	0.0694	1.511	0.04	0.5160	0.379	1.56	0.9406	0.026
-2.96	0.0015	2.960	-1.44	0.0749	1.474	0.08	0.5319	0.360	1.60	0.9452	0.023
-2.92	0.0018	2.921	-1.40	0.0808	1.437	0.12	0.5478	0.342	1.64	0.9495	0.021
-2.88	0.0020	2.881	-1.36	0.0869	1.400	0.16	0.5636	0.324	1.68	0.9535	0.019
-2.84	0.0023	2.841	-1.32	0.0934	1.364	0.20	0.5793	0.307	1.72	0.9573	0.017
-2.80	0.0026	2.801	-1.28	0.1003	1.327	0.24	0.5948	0.290	1.76	0.9608	0.016
-2.76	0.0029	2.761	-1.24	0.1075	1.292	0.28	0.6103	0.274	1.80	0.9641	0.014
-2.72	0.0033	2.721	-1.20	0.1151	1.256	0.32	0.6255	0.259	1.84	0.9671	0.013
-2.68	0.0037	2.681	-1.16	0.1230	1.221	0.36	0.6406	0.245	1.88	0.9699	0.012
-2.64	0.0041	2.641	-1.12	0.1314	1.186	0.40	0.6554	0.230	1.92	0.9726	0.010
-2.60	0.0047	2.601	-1.08	0.1401	1.151	0.44	0.6700	0.217	1.96	0.9750	0.009
-2.56	0.0052	2.562	-1.04	0.1492	1.117	0.48	0.6844	0.204	2.00	0.9772	0.008
-2.52	0.0059	2.522	-1.00	0.1587	1.083	0.52	0.6985	0.192	2.04	0.9793	0.008
-2.48	0.0066	2.482	-0.96	0.1685	1.050	0.56	0.7123	0.180	2.08	0.9812	0.007
-2.44	0.0073	2.442	-0.92	0.1788	1.017	0.60	0.7257	0.169	2.12	0.9830	0.006
-2.40	0.0082	2.403	-0.88	0.1894	0.984	0.64	0.7389	0.158	2.16	0.9846	0.005
-2.36	0.0091	2.363	-0.84	0.2005	0.952	0.68	0.7517	0.148	2.20	0.9861	0.005
-2.32	0.0102	2.323	-0.80	0.2119	0.920	0.72	0.7642	0.138	2.24	0.9875	0.004
-2.28	0.0113	2.284	-0.76	0.2236	0.889	0.76	0.7764	0.129	2.28	0.9887	0.004
-2.24	0.0125	2.244	-0.72	0.2358	0.858	0.80	0.7881	0.120	2.32	0.9898	0.003
-2.20	0.0139	2.205	-0.68	0.2483	0.828	0.84	0.7995	0.112	2.36	0.9909	0.003
-2.16	0.0154	2.165	-0.64	0.2611	0.798	0.88	0.8106	0.104	2.40	0.9918	0.003
-2.12	0.0170	2.126	-0.60	0.2743	0.769	0.92	0.8212	0.097	2.44	0.9927	0.002
-2.08	0.0188	2.087	-0.56	0.2877	0.740	0.96	0.8315	0.090	2.48	0.9934	0.002
-2.04	0.0207	2.048	-0.52	0.3015	0.712	1.00	0.8413	0.083	2.52	0.9941	0.002
-2.00	0.0228	2.008	-0.48	0.3156	0.684	1.04	0.8508	0.077	2.56	0.9948	0.002
-1.96	0.0250	1.969	-0.44	0.3300	0.657	1.08	0.8599	0.071	2.60	0.9953	0.001
-1.92	0.0274	1.930	-0.40	0.3446	0.630	1.12	0.8686	0.066	2.64	0.9959	0.001
-1.88	0.0301	1.892	-0.36	0.3594	0.605	1.16	0.8770	0.061	2.68	0.9963	0.001
-1.84	0.0329	1.853	-0.32	0.3745	0.579	1.20	0.8849	0.056	2.72	0.9967	0.001
-1.80	0.0359	1.814	-0.28	0.3897	0.554	1.24	0.8925	0.052	2.76	0.9971	0.001
-1.76	0.0392	1.776	-0.24	0.4052	0.530	1.28	0.8997	0.047	2.80	0.9974	0.001
-1.72	0.0427	1.737	-0.20	0.4207	0.507	1.32	0.9066	0.044	2.84	0.9977	0.001
-1.68	0.0465	1.699	-0.16	0.4364	0.484	1.36	0.9131	0.040	2.88	0.9980	0.001
-1.64	0.0505	1.661	-0.12	0.4522	0.462	1.40	0.9192	0.037	2.92	0.9982	0.001
-1.60	0.0548	1.623	-0.08	0.4681	0.440	1.44	0.9251	0.034	2.96	0.9985	0.000
-1.56	0.0594	1.586	-0.04	0.4840	0.419	1.48	0.9306	0.031	3.00	0.9987	0.000
-1.52	0.0643	1.548	0.00	0.5000	0.399	1.52	0.9357	0.028			

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Newsvendor: Bread

$\mu_X = 300$
$\sigma_X = 50$

$F(q^*) = 0.50$

$z = \dfrac{x - \mu_X}{\sigma_X} \quad \Rightarrow x = \mu_X + z\sigma_X$

$F(q^*) = 0.50 = \Phi(z) \;\Rightarrow z^* = 0$

$q^* = \mu_X + z^*\sigma_X = 300 + (0)(50) = 300 \text{ units}$

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Newsvendor: Bread Take 2

A supermarket must decide how many loaves of a particular type of whole wheat bread to purchase every day. The daily demand for this type of bread is approximately normally distributed with a mean of 300 and a standard deviation of 50 loaves.

A loaf of bread sells for $2.89 and costs the supermarket $1.59.

A loaf of bread that is not sold by the end of the day can be sold as a *"day old"* item the next day for $0.99.

How many loaves of this type of whole wheat bread should the supermarket purchase every day?

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Newsvendor: Bread Take 2

$cost = \$1.59/\text{unit}$
$selling\ price = \$2.89/\text{unit}$
$salvage\ revenue = \$0.99/\text{unit}$

- Unit underage cost

$$c_u = 2.89 - 1.59 = 1.30/\text{unit}$$

- Unit overage cost

$$c_o = 1.59 - 0.99 = 0.60/\text{unit}$$

- Critical ratio

$$F(q^*) = \frac{c_u}{c_u + c_0} = \frac{1.3}{1.3 + 0.6} = 0.6842$$

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Z-table

Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)
-3.00	0.0013	3.000	-1.48	0.0694	1.511	0.04	0.5160	0.379	1.56	0.9406	0.026
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-2.68	0.0037	2.681	-1.16	0.1230	1.221	0.36	0.6406	0.245	1.88	0.9699	0.012
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-2.60	0.0047	2.601	-1.08	0.1401	1.151	0.44	0.6700	0.217	1.96	0.9750	0.009
-2.56	0.0052	2.562	-1.04	0.1492	1.117	0.48	0.6844	0.204	2.00	0.9772	0.008
-2.52	0.0059	2.522	-1.00	0.1587	1.083	0.52	0.6985	0.192	2.04	0.9793	0.008
-2.48	0.0066	2.482	-0.96	0.1685	1.050	0.56	0.7123	0.180	2.08	0.9812	0.007
-2.44	0.0073	2.442	-0.92	0.1788	1.017	0.60	0.7257	0.169	2.12	0.9830	0.006
-2.40	0.0082	2.403	-0.88	0.1894	0.984	0.64	0.7389	0.158	2.16	0.9846	0.005
-2.36	0.0091	2.363	-0.84	0.2005	0.952	0.68	0.7517	0.148	2.20	0.9861	0.005
-2.32	0.0102	2.323	-0.80	0.2119	0.920	0.72	0.7642	0.138	2.24	0.9875	0.004
-2.28	0.0113	2.284	-0.76	0.2236	0.889	0.76	0.7764	0.129	2.28	0.9887	0.004
-2.24	0.0125	2.244	-0.72	0.2358	0.858	0.80	0.7881	0.120	2.32	0.9898	0.003
-2.20	0.0139	2.205	-0.68	0.2483	0.828	0.84	0.7995	0.112	2.36	0.9909	0.003
-2.16	0.0154	2.165	-0.64	0.2611	0.798	0.88	0.8106	0.104	2.40	0.9918	0.003
-2.12	0.0170	2.126	-0.60	0.2743	0.769	0.92	0.8212	0.097	2.44	0.9927	0.002
-2.08	0.0188	2.087	-0.56	0.2877	0.740	0.96	0.8315	0.090	2.48	0.9934	0.002
-2.04	0.0207	2.048	-0.52	0.3015	0.712	1.00	0.8413	0.083	2.52	0.9941	0.002
-2.00	0.0228	2.008	-0.48	0.3156	0.684	1.04	0.8508	0.077	2.56	0.9948	0.002
-1.96	0.0250	1.969	-0.44	0.3300	0.657	1.08	0.8599	0.071	2.60	0.9953	0.001
-1.92	0.0274	1.930	-0.40	0.3446	0.630	1.12	0.8686	0.066	2.64	0.9959	0.001
-1.88	0.0301	1.892	-0.36	0.3594	0.605	1.16	0.8770	0.061	2.68	0.9963	0.001
-1.84	0.0329	1.853	-0.32	0.3745	0.579	1.20	0.8849	0.056	2.72	0.9967	0.001
-1.80	0.0359	1.814	-0.28	0.3897	0.554	1.24	0.8925	0.052	2.76	0.9971	0.001
-1.76	0.0392	1.776	-0.24	0.4052	0.530	1.28	0.8997	0.047	2.80	0.9974	0.001
-1.72	0.0427	1.737	-0.20	0.4207	0.507	1.32	0.9066	0.044	2.84	0.9977	0.001
-1.68	0.0465	1.699	-0.16	0.4364	0.484	1.36	0.9131	0.040	2.88	0.9980	0.001
-1.64	0.0505	1.661	-0.12	0.4522	0.462	1.40	0.9192	0.037	2.92	0.9982	0.001
-1.60	0.0548	1.623	-0.08	0.4681	0.440	1.44	0.9251	0.034	2.96	0.9985	0.000
-1.56	0.0594	1.586	-0.04	0.4840	0.419	1.48	0.9306	0.031	3.00	0.9987	0.000
-1.52	0.0643	1.548	0.00	0.5000	0.399	1.52	0.9357	0.028			

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Newsvendor: Bread Take 2

$\mu_X = 300$
$\sigma_X = 50$

$F(q^*) = 0.6842$

$z = \dfrac{x - \mu_X}{\sigma_X} \quad \Rightarrow x = \mu_X + z\sigma_X$

$F(q^*) = 0.6842 = \Phi(z) \Rightarrow z^* = 0.4795$

$q^* = \mu_X + z^*\sigma_X = 300 + (0.4795)(50) = 323.98 \text{ units}$

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Newsvendor: So far…

- Consider a cost minimization objective with unit cost of underage and unit cost of overage…

- How do these map to problem parameters we can identify in real life?

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Newsvendor: Profit maximization: Additional notation

p unit selling price (\$/unit);

c unit purchase (or procurement) cost (\$/unit);

v unit salvage revenue (\$/unit);

b unit loss of goodwill(\$/unit);

$p(q, X)$ total net profit at the end of the period (\$);

$P(q) = [p(q, X)]$ the expected net profit function (\$).

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The loss-of-goodwill cost: b

- What is it?

- How do we quantify it?

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Newsvendor: Profit maximization

What is the realized net profit if the order quantity is q and the random demand is X?

$$p(q,X) = \begin{cases} -cq + pX + v(q-X) & X \le q \\ -cq + pq - b(X-q) & q < X \end{cases}$$

What is the expected total cost over all possible values of X?

$$E_X[p(q,X)] = \int_0^q [-cq + pX + v(q-X)]f(x)dx + \int_q^\infty [-cq + pq - b(X-q)]f(x)dx$$

$$= P(q)$$

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Newsvendor: Profit maximization

$$p(q,X) = \begin{cases} -cq + pX + v(q-X) & X \le q \\ -cq + pq - b(X-q) & q < X \end{cases}$$

$$P(q) = -cq\int_0^q f(x)dx + p\int_0^q xf(x)dx + v\int_0^q (q-x)f(x)dx$$
$$-cq\int_q^\infty f(x)dx + p\int_q^\infty qf(x)dx - b\int_q^\infty (x-q)f(x)dx$$

Add and subtract the term $p\int_q^\infty xf(x)dx$

$$P(q) = \boxed{-cq\int_0^q f(x)dx} + \boxed{p\int_0^q xf(x)dx} + v\int_0^q (q-x)f(x)dx$$
$$\boxed{-cq\int_q^\infty f(x)dx} + p\boxed{\int_q^\infty qf(x)dx} - b\int_q^\infty (x-q)f(x)dx +$$
$$+\boxed{p\int_q^\infty xf(x)dx} - p\boxed{\int_q^\infty xf(x)dx}$$

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Newsvendor: Profit maximization

$$P(q) = \boxed{-cq\left[\int_0^q f(x)dx + \int_q^\infty f(x)dx\right]} + \boxed{p\left[\int_0^q xf(x)dx + \int_q^\infty xf(x)dx\right]}$$
$$+v\int_0^q (q-x)f(x)dx - \boxed{p\int_q^\infty (x-q)f(x)dx} - \boxed{b\int_q^\infty (x-q)f(x)dx}$$

$$P(q) = -cq + pE[X] + v\int_0^q (q-x)f(x)dx - \boxed{(b+p)\int_q^\infty (x-q)f(x)dx}$$

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Newsvendor: Profit maximization: First order condition

$$P(q) = -cq + pE[X] + v\int_0^q (q-x)f(x)dx - (b+p)\int_q^\infty (x-q)f(x)dx$$

$$\frac{dP(q)}{dq} = -c \ +0 \ +v\left[\int_0^q (1)f(x)dx + (q-q)(1) - (q-0)(0)\right]$$

$$-(b+p)\left[\int_q^\infty (-1)f(x)dx + (\infty-q)(0) - (q-q)(1)\right]$$

$$= -c + v\int_0^q f(x)dx + (b+p)\int_q^\infty f(x)dx$$

$$= -c + vF(q) + (b+p)[1-F(q)]$$

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Newsvendor: Profit maximization: Second order condition

$$\frac{dP(q)}{dq} = -c + vF(q) + (b+p)[1 - F(q)]$$

$$\frac{d^2P(q)}{dq^2} = vf(q) - (b+p)f(q) < 0$$

$$\frac{dP(q)}{dq} = -c + vF(q) + (b+p)[1 - F(q)] = 0$$

$$F(q^*) = \frac{p - c + b}{b + p - v}$$

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Newsvendor: A comparison

COST MINIMIZATION

- Unit cost of having too much inventory (c_o) and unit cost of having not enough inventory (c_u)

$$F(q^*) = \frac{c_u}{c_u + c_o}$$

PROFIT MAXIMIZATION

- Unit selling price (p), unit procurement cost (c), unit loss-of-goodwill cost (b) and unit salvage revenue (v)

$$F(q^*) = \frac{p - c + b}{p + b - v} = \frac{c_u}{c_u + c_o}$$

$$c_u = p - c + b$$

$$c_o = c - v$$

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Newsvendor: Florida Gator Shirts

Kohl's sells Florida Gators Gear. Gainesville store manager must decide how many polo shirts to purchase for 2021 Football Season. Historically, the demand for polo shirts is approximately normally distributed with a mean of 3,000 and a standard deviation of 400 units.

A shirt sells for $29.99 and costs the store $12.59. The store estimates the loss-of-goodwill cost to be $10/unit. A shirt that is not sold by the end of the season can easily sold as a *"past season"* shirt at the end of the season for $9.99.

How many of these polo shirts should the store manager order prior to the 2021 football season?

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Newsvendor: Florida Gator shirts

$c = \$12.59/\text{unit}$
$p = \$29.99/\text{unit}$
$v = \$9.99/\text{unit}$
$b = \$10.00/\text{unit}$

- Unit underage cost

$$c_u = p - c + b = 29.99 - 12.59 + 10 = \$27.40/\text{shirt}$$

- Unit overage cost

$$c_o = c - v = 12.59 - 9.99 = \$2.60/\text{shirt}$$

- Critical ratio

$$F(q^*) = \frac{27.40}{27.40 + 2.60} = \frac{27.40}{30} = 0.9133$$

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Z-table

Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)
-3.00	0.0013	3.000	-1.48	0.0694	1.511	0.04	0.5160	0.379	1.56	0.9406	0.026
-2.96	0.0015	2.960	-1.44	0.0749	1.474	0.08	0.5319	0.360	1.60	0.9452	0.023
-2.92	0.0018	2.921	-1.40	0.0808	1.437	0.12	0.5478	0.342	1.64	0.9495	0.021
-2.88	0.0020	2.881	-1.36	0.0869	1.400	0.16	0.5636	0.324	1.68	0.9535	0.019
-2.84	0.0023	2.841	-1.32	0.0934	1.364	0.20	0.5793	0.307	1.72	0.9573	0.017
-2.80	0.0026	2.801	-1.28	0.1003	1.327	0.24	0.5948	0.290	1.76	0.9608	0.016
-2.76	0.0029	2.761	-1.24	0.1075	1.292	0.28	0.6103	0.274	1.80	0.9641	0.014
-2.72	0.0033	2.721	-1.20	0.1151	1.256	0.32	0.6255	0.259	1.84	0.9671	0.013
-2.68	0.0037	2.681	-1.16	0.1230	1.221	0.36	0.6406	0.245	1.88	0.9699	0.012
-2.64	0.0041	2.641	-1.12	0.1314	1.186	0.40	0.6554	0.230	1.92	0.9726	0.010
-2.60	0.0047	2.601	-1.08	0.1401	1.151	0.44	0.6700	0.217	1.96	0.9750	0.009
-2.56	0.0052	2.562	-1.04	0.1492	1.117	0.48	0.6844	0.204	2.00	0.9772	0.008
-2.52	0.0059	2.522	-1.00	0.1587	1.083	0.52	0.6985	0.192	2.04	0.9793	0.008
-2.48	0.0066	2.482	-0.96	0.1685	1.050	0.56	0.7123	0.180	2.08	0.9812	0.007
-2.44	0.0073	2.442	-0.92	0.1788	1.017	0.60	0.7257	0.169	2.12	0.9830	0.006
-2.40	0.0082	2.403	-0.88	0.1894	0.984	0.64	0.7389	0.158	2.16	0.9846	0.005
-2.36	0.0091	2.363	-0.84	0.2005	0.952	0.68	0.7517	0.148	2.20	0.9861	0.005
-2.32	0.0102	2.323	-0.80	0.2119	0.920	0.72	0.7642	0.138	2.24	0.9875	0.004
-2.28	0.0113	2.284	-0.76	0.2236	0.889	0.76	0.7764	0.129	2.28	0.9887	0.004
-2.24	0.0125	2.244	-0.72	0.2358	0.858	0.80	0.7881	0.120	2.32	0.9898	0.003
-2.20	0.0139	2.205	-0.68	0.2483	0.828	0.84	0.7995	0.112	2.36	0.9909	0.003
-2.16	0.0154	2.165	-0.64	0.2611	0.798	0.88	0.8106	0.104	2.40	0.9918	0.003
-2.12	0.0170	2.126	-0.60	0.2743	0.769	0.92	0.8212	0.097	2.44	0.9927	0.002
-2.08	0.0188	2.087	-0.56	0.2877	0.740	0.96	0.8315	0.090	2.48	0.9934	0.002
-2.04	0.0207	2.048	-0.52	0.3015	0.712	1.00	0.8413	0.083	2.52	0.9941	0.002
-2.00	0.0228	2.008	-0.48	0.3156	0.684	1.04	0.8508	0.077	2.56	0.9948	0.002
-1.96	0.0250	1.969	-0.44	0.3300	0.657	1.08	0.8599	0.071	2.60	0.9953	0.001
-1.92	0.0274	1.930	-0.40	0.3446	0.630	1.12	0.8686	0.066	2.64	0.9959	0.001
-1.88	0.0301	1.892	-0.36	0.3594	0.605	1.16	0.8770	0.061	2.68	0.9963	0.001
-1.84	0.0329	1.853	-0.32	0.3745	0.579	1.20	0.8849	0.056	2.72	0.9967	0.001
-1.80	0.0359	1.814	-0.28	0.3897	0.554	1.24	0.8925	0.052	2.76	0.9971	0.001
-1.76	0.0392	1.776	-0.24	0.4052	0.530	1.28	0.8997	0.047	2.80	0.9974	0.001
-1.72	0.0427	1.737	-0.20	0.4207	0.507	1.32	0.9066	0.044	2.84	0.9977	0.001
-1.68	0.0465	1.699	-0.16	0.4364	0.484	1.36	0.9131	0.040	2.88	0.9980	0.001
-1.64	0.0505	1.661	-0.12	0.4522	0.462	1.40	0.9192	0.037	2.92	0.9982	0.001
-1.60	0.0548	1.623	-0.08	0.4681	0.440	1.44	0.9251	0.034	2.96	0.9985	0.000
-1.56	0.0594	1.586	-0.04	0.4840	0.419	1.48	0.9306	0.031	3.00	0.9987	0.000
-1.52	0.0643	1.548	0.00	0.5000	0.399	1.52	0.9357	0.028			

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Newsvendor: Florida Gator Shirts

$\mu_X = 3{,}000$
$\sigma_X = 400$

$$z = \frac{x - \mu_X}{\sigma_X} \quad \Rightarrow x = \mu_X + z\sigma_X$$

$$F(q^*) = 0.9133 = \Phi(z)$$

1.36	$F(1.36) = 0.9131$
$z^* = ?$	$F(q^*) = 0.9133$
1.40	$F(1.40) = 0.9192$

$$\frac{z^* - 1.36}{0.9133 - 0.9131} = \frac{1.40 - 1.36}{0.9192 - 0.9131}$$

$$\Rightarrow z^* = 1.36 + (0.9133 - 0.9131)\frac{1.40 - 1.36}{0.9192 - 0.9131} = 1.3613$$

$$q^* = \mu_X + z^*\sigma_X = 3{,}000 + (1.3613)(400) = 3{,}544.52 \text{ shirts}$$

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Longshoreman strike at East and Gulf Coast ports…Update…

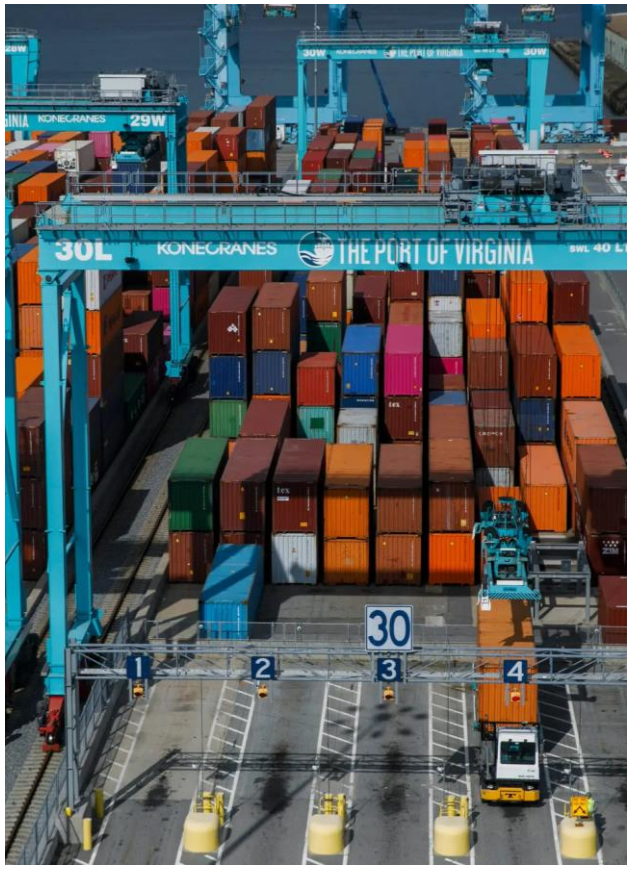
- *…There is no practical alternative to the East and Gulf Coast ports for moving many goods in and out of the eastern half of the country. And the ports can't operate without longshoremen, who move metal boxes called containers on and off ships and handle other economically crucial cargo, like cars and heavy machinery.*

- *Fearing a strike, many businesses ordered goods early so that they cleared the East and Gulf Coast ports before Tuesday. But much cargo didn't make it through in time and could be stranded at sea. Even if a strike is short, it could take weeks to clear the backlog.*

- *John Wrenn, the chief operating officer at MHW, an alcoholic beverages distributor based in Manhasset, N.Y., is concerned that a strike might delay shipments and leave his business short in the lead up to Thanksgiving.*

- *"Those are big opportunities for sales that will just be lost because products will not be on the shelves," he said. MHW imports over 2,000 containers a year through the Port of New York and New Jersey, the third busiest port in the United States.*

- *Stefanie Katzman, executive vice president at S. Katzman Produce, a fruit and vegetable supplier in the Bronx, is particularly concerned that mangos, which are imported from Brazil, will spoil if they are stuck on a ship. She is considering flying some mangos in, a much more expensive option. "There's nowhere near as much room to fly product as there is to send it in by boat," Ms. Katzman said.*



https://www.nytimes.com/2024/09/30/business/port-strike-dockworkers.html

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: STOCHASTIC DEMAND FINITE HORIZON – MULTIPLE PERIODS

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Newsvendor: So far…

- Consider a cost minimization/profit maximization objective…

- What if the items left over at the end of the period did not perish quickly and can be used to satisfy demand in subsequent periods?

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Newsvendor: Multiple periods

- Handling of excess demand has an impact on the analysis
  - Full backlogging
  - Lost sales

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Newsvendor: Multiple periods: Full backlogging

- Let X_t denote demand in period t such that all X_t's are independent and identically distributed (iid) with pdf $f(x)$ and cdf $F(x)$

- Since demand in each period is iid, all periods should have identical initial inventory levels entering each period.

- Let q denote the *order-up-to level*, and T denote the *finite horizon length*.

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Let's play a game

Let $q = 100$ units and $T = 5$.

Period	Initial Inventory	Observed Demand	Ending Inventory	Order Quantity
0				100
1	100	70	30	70
2	100	90	10	90
3	100	110	-10	110
4	100	95	5	95
5	100	101	-1	0

The quantity ordered at the end of any period t, such that $0 < t < T$, is equal to the observed demand in the period.

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Newsvendor: Multiple periods: Full backlogging

How much do we order in each period?

- In period 0, to initialize the system, we place an order for q units.

- In any period t for $t = 1, \ldots T - 1$, we observe the demand to be X_t. In any period t, to restore the inventory level up to q for period $t + 1$, we have to place an order of X_t units.

- In period T, however, we place no new order since after the end of the planning horizon there is no demand to satisfy.

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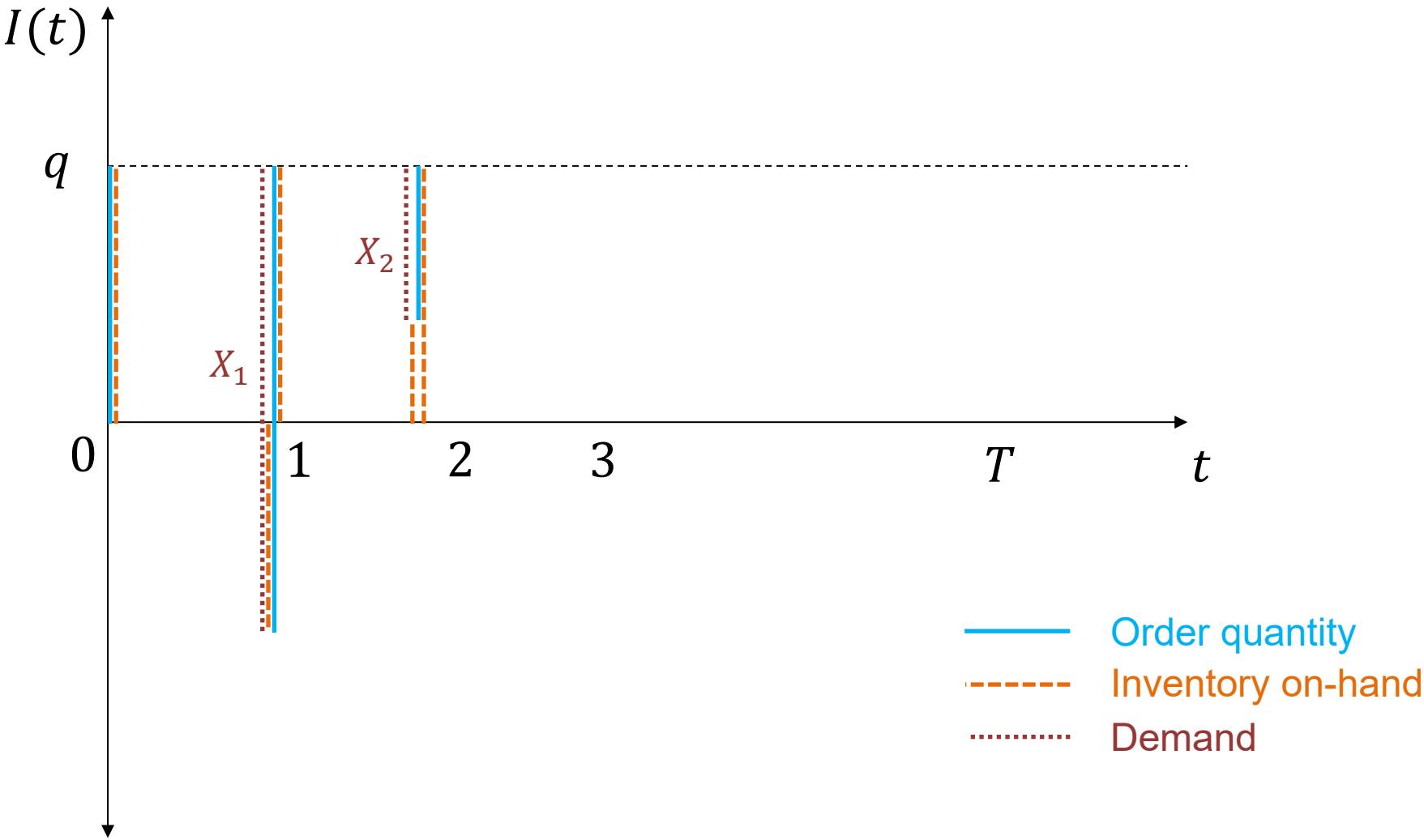
Newsvendor: Multiple periods: Full backlogging



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Newsvendor: Multiple periods: Full backlogging

What is the purchase cost for period 1?

$$-cq$$

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Newsvendor: Multiple periods: Full backlogging

What is the purchase cost for periods 2 through T?

$$-cE[X_1 + X_2 + \cdots + X_{T-1}] = -c\left[E[X_1] + E[X_2] + \cdots + E[X_{T-1}]\right]$$

$$= -c(T-1)\mu$$

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Newsvendor: Multiple periods: Full backlogging

What is the sales revenue in periods 1 through $T-1$?

$$pE[X_1 + X_2 + \cdots + X_{T-1}] = p\left[E[X_1] + E[X_2] + \cdots + E[X_{T-1}]\right]$$

$$= p(T-1)\mu$$

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Newsvendor: Multiple periods: Full backlogging

What is the sales revenue in period T?

$$pE[\min\{q, X_T\}] = p\left[\int_0^q xf(x)dx + \int_q^\infty qf(x)dx\right]$$

$$= p\left[\int_0^q xf(x)dx + \int_q^\infty xf(x)dx - \int_q^\infty xf(x)dx + \int_q^\infty qf(x)dx\right]$$

$$= p\left[\mu - \int_q^\infty (x-q)f(x)dx\right]$$

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Newsvendor: Multiple periods: Full backlogging

What is the expected backordering or inventory holding costs in periods 1 through T?

$$-T\left\{h\int_0^q (q-x)f(x)dx + b\int_q^\infty (x-q)f(x)dx\right\} = -T\mathcal{L}(q)$$

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Newsvendor: Multiple periods: Full backlogging

Consequently, the expected net profit throughout the planning horizon is given by

$$-cq + (p-c)(T-1)\mu + p\left[\mu - \int_q^\infty (x-q)f(x)dx\right] - T\mathcal{L}(q)$$

And, the expected net profit throughout the planning horizon is given by

$$\frac{-cq + (p-c)(T-1)\mu + p\left[\mu - \int_q^\infty (x-q)f(x)dx\right] - T\mathcal{L}(q)}{T}$$

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Newsvendor: Multiple periods: Full backlogging

Let us examine the behavior of the average expected net profit as $T \to \infty$

$$\lim_{T\to\infty} \frac{\boxed{-cq} + \boxed{(p-c)(T-1)\mu} + \boxed{p\left[\mu - \int_q^\infty (x-q)f(x)dx\right]} - \boxed{T\mathcal{L}(q)}}{T}$$

$$=$$

$$\equiv P(q)$$

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Newsvendor: Multiple periods: Full backlogging

First order condition of $P(q)$

$$\frac{dP(q)}{dq} = -\mathcal{L}'(q) \ = 0$$

$$F(q^*) = \frac{b}{b+h}$$

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Example

Just Around the Corner is a small bookstore in Utopia. Books are replenished weekly from a publishing distributor. Copies of the books that are not sold at the end of a week are still kept on the shelves for future sales.

Since Just Around the Corner is the only bookstore in Utopia, if a book is out of stock, then the customers request copies of the book and are willing to wait until the following week.

Walter Isaacson's biography of Steve Jobs sells for $35.00 and cost the bookstore $11.95 per book.

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Example: Full backlogging

The owner of Just Around the Corner estimates a loss-of-goodwill cost of $4 each time a book is back-ordered.

Weekly demand for Steve Jobs's biography can be approximated by a normal distribution with a mean of 20 and a standard deviation of 8 books per week.

The owner uses a weekly interest rate of 5 percent to determine the inventory holding cost. How many copies of the book should be purchased at the beginning of each week?

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Example: Full backlogging

Given

$$b = \$4/\text{unit}$$

$$h = ic = (0.05)(11.95) = \$0.5975/\text{unit}$$

the critical ratio is given by

$$F(q^*) = \frac{b}{b+h} = \frac{4}{4+0.5975} = 0.87$$

since the demand is normally distributed, corresponding z value is given by

$$\Phi^{-1}(0.87) = 1.1264$$

Consequently, we have

$$q^* = \mu + z^*\sigma = 20 + (1.1264)(8) = 29.0112 \text{ books}$$

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Newsvendor: Multiple periods

- Handling of excess demand has an impact on the analysis
  - Full backlogging
  - Lost sales

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Newsvendor: Multiple periods: Lost sales

- Let X_t denote demand in period t such that all X_t's are independent and identically distributed (iid) with pdf $f(x)$ and cdf $F(x)$

- Since demand in each period is iid, all periods should have identical initial inventory levels entering each period.

- Let q denote the *order-up-to level*, and T denote the *finite horizon length*.

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Let's play a game

Let $q = 100$ units and $T = 5$.

Period	Initial Inventory	Observed Demand	Ending Inventory	Order Quantity
0				100
1	100	70	30	70
2	100	51	49	51
3	100	120	0	100
4	100	90	10	90
5	100	66	34	0

The order quantity is equal to the minimum of the order up to level and the Observed demand in any period.

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Newsvendor: Multiple periods: Lost sales

How much do we order in each period?

- In period 0, to initialize the system, we place an order for q units.

- In any period t for $t = 1, \ldots T - 1$, we observe the demand to be X_t. In any period t, to restore the inventory level up to q for period $t + 1$, we have to place an order of $\min\{q, X_t\}$units.

- In period T, we place no new order since after the end of the planning horizon there is no demand to satisfy.

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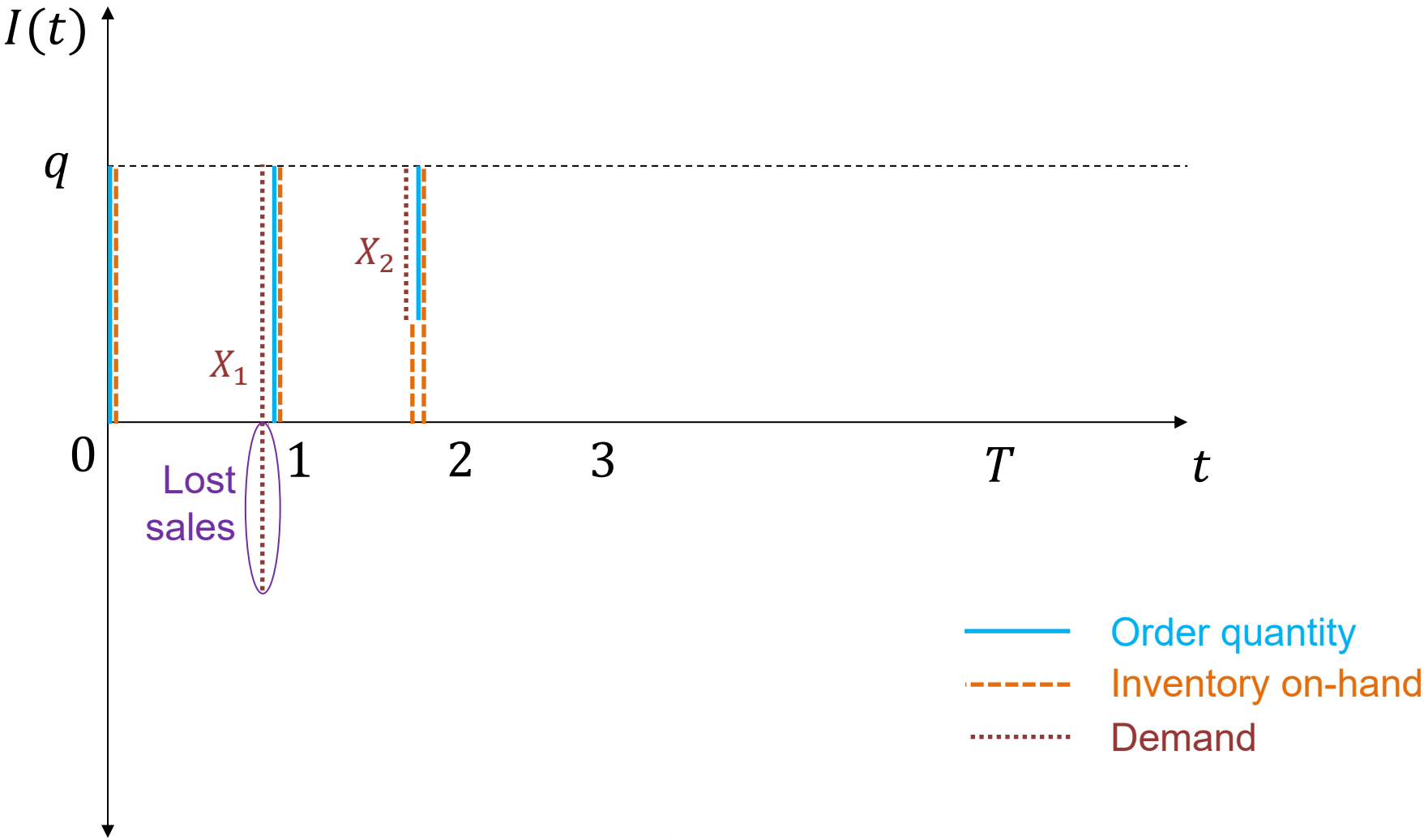
Newsvendor: Multiple periods: Lost sales



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Newsvendor: Multiple periods: Lost sales

What is the purchase cost for period 1?

$-cq$

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Newsvendor: Multiple periods: Lost sales

What is the purchase cost for periods 2 through T?

Let S_t be a random variable that represents the amount sold in period t

$$-cE[S_1 + S_2 + \cdots + S_{T-1}] = -c\left[E[S_1] + E[S_2] + \cdots + E[S_{T-1}]\right]$$

$$= -c(T-1)E[S_1]$$

$$E[S_1] = E[\min\{q, X_1\}]$$

$$= \int_0^q xf(x)dx + \int_q^\infty qf(x)dx$$

$$= \int_0^q xf(x)dx + \int_q^\infty xf(x)dx - \int_q^\infty xf(x)dx + \int_q^\infty qf(x)dx$$

$$= \mu - \int_q^\infty (x-q)f(x)dx$$

We have done this before!! Recall slide 56…

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Newsvendor: Multiple periods: Lost sales

What is the sales revenue in periods 1 through T?

$$pE[S_1 + S_2 + \cdots + S_T] = p\left[E[S_1] + E[S_2] + \cdots + E[S_T]\right]$$

$$= pTE[S_1]$$

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Newsvendor: Multiple periods: Lost sales

What is the expected backordering or inventory holding costs in periods 1 through T?

$$-T\left\{h\int_0^q (q-x)f(x)dx + b\int_q^\infty (x-q)f(x)dx\right\} = -T\mathcal{L}(q)$$

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Newsvendor: Multiple periods: Lost sales

Consequently, the expected net profit throughout the planning horizon is given by

$$-cq - c(T-1)E[S_1] + pTE[S_1] - T\mathcal{L}(q)$$

and the expected net profit throughout the planning horizon is given by

From slides single period newsvendor and the last period of multi-period newsvendor with full backlogging

$$\frac{-cq - [c(T-1) + pT]\left[\mu - \int_q^\infty (x-q)f(x)dx\ \right] - T\mathcal{L}(q)}{T}$$

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Newsvendor: Multiple periods: Lost sales

Let us examine the behavior of the average expected net profit as $T \to \infty$

$$\lim_{T\to\infty} \frac{\boxed{-cq} + \boxed{[-c(T-1)+pT]\left[\mu - \int_q^\infty (x-q)f(x)dx\right]} - \boxed{T\mathcal{L}(q)}}{T}$$

$$= (p-c)\left[\mu - \int_q^\infty (x-q)f(x)dx\right] - \mathcal{L}(q)$$

$$\equiv P(q)$$

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Newsvendor: Multiple periods: Lost sales

First order condition of $P(q)$

$$\frac{dP(q)}{dq} = 0$$

$$F(q^*) = \frac{p + b - c}{p + b - c + h}$$

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Example: Lost sales

Recall the previous example. Suppose that a book retailer, Barns, decided to enter the Utopian market and opened a store one block away. Now if Just Around the Corner is out of stock, the customers go and buy a copy from Barns.

In this situation, how many copies of Steve Jobs' biography should the owner of Just Around the Corner order each week?

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Example: Lost sales

Given

$$b = \$4/\text{unit}$$
$$h = ic = (0.05)(11.95) = \$0.5975/\text{unit}$$
$$p = \$35/\text{unit}$$
$$c = \$11.95/\text{unit}$$

the critical ratio is given by

$$F(q^*) = \frac{p + b - c}{p + b - c + h} = \frac{35 + 4 - 11.95}{35 + 4 - 11.95 + 0.5975} = 0.9784$$

since the demand is normally distributed, corresponding z value is given by

$$\Phi^{-1}(0.9784) = 2.0218$$

Consequently, we have

$$q^* = \mu + z^*\sigma = 20 + (2.0218)(8) \quad = 36.1744 \text{ books}$$

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Epilogue

Full backlogging

Optimal order quantity is

29.0112 books

Lost sales

Optimal order quantity is

36.1744 books

Does this make sense? That is, does it make sense that the optimal order quantity is <u>larger</u> when excess demand is lost as opposed to it being backlogged?

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